SOLEIL DU MATIN

Coffee & Bakery Sales Performance Dashboard

An executive Excel dashboard case study that turns 149,116 raw transactions into a one-page decision tool.

Power Query · Power Pivot · DAX · Pivot Tables · VBA

Built using Microsoft Excel 365



One-page executive dashboard · January – June 2023 · three New York locations

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$698.81K	149.1K	3	18	6 mo
TOTAL REVENUE	TRANSACTIONS	STORE LOCATIONS	DAX MEASURES	JAN – JUN 2023

1 Project Overview

Soleil Du Matin is a coffee and bakery brand with three locations in New York: Hell's Kitchen, Astoria, and Lower Manhattan. For this project, I built a one-page executive sales dashboard in Microsoft Excel covering January to June 2023. The goal was to give management a clear view of how the business is performing and where they should focus next.

The dashboard shows revenue trends, store comparisons, customer traffic by hour and day, product category breakdown, and a ranked list of top products. It has slicers for month and store location, and every chart, KPI card, and ranking updates automatically whenever a filter changes.

The Brand Story

I created the coffee shop brand from scratch, including the business concept, name, and logo. "Soleil du Matin" means "Morning Sun" in French. I chose the name because coffee is closely associated with the start of the day, making the morning sun a fitting symbol of warmth, energy, and new beginnings.

The name also carries a personal meaning. "Soleil" is my daughter's name, which inspired the brand and made the project more meaningful to me. I incorporated warm, sunrise-inspired colors throughout the logo and dashboard to create a consistent brand identity, from the visuals to the analytics.

2 Business Problem

The company had six months of raw transaction data spread across three stores, but no easy way to answer basic questions like "Are we hitting our target?", "Which store is winning?", or "When should we schedule more staff?" Management needed one page they could open, read in a few minutes, and use to make decisions on staffing, promotions, and product focus.

3 Business Questions

The dashboard was built to answer these questions:

1. Is revenue growing, and will the business hit its $700K target for the first half of the year?
2. Which store earns the most, and how far behind are the others?
3. What times and days do customers actually visit?
4. Which categories and products bring in the most revenue?
5. Is the average order value meeting the $5 target?
6. Where are the biggest opportunities for staffing, promotions, and product focus?

4 Data Set

Source: A raw CSV file (`Coffee Shop Sales.csv`) with transaction-level sales data for January to June 2023.

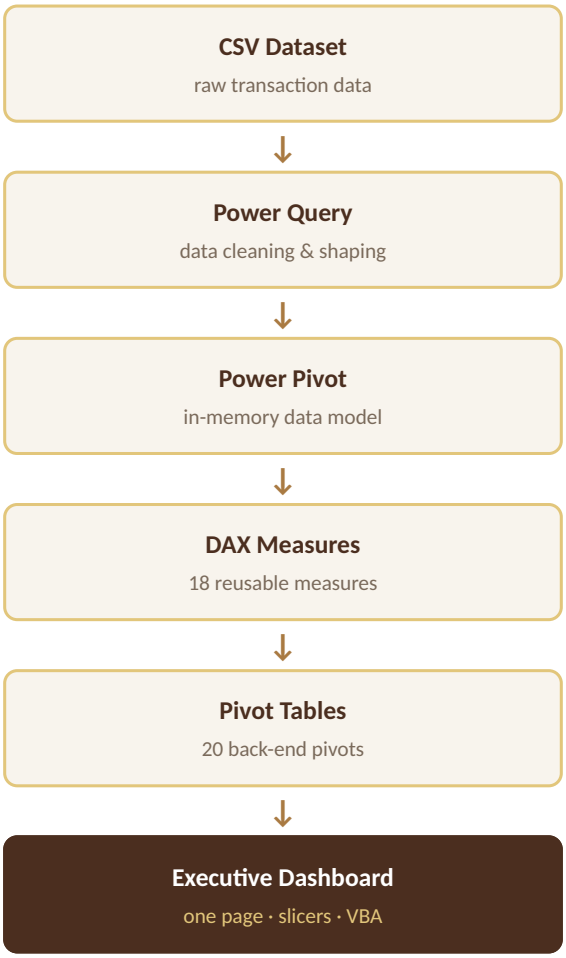
Size: 149,116 transactions across the three store locations.

Original columns (11): `transaction_id`, `transaction_date`, `transaction_time`, `transaction_qty`, `store_id`, `store_location`, `product_id`, `unit_price`, `product_category`, `product_type`, `product_detail`.

Data Overview

The data covers 6 months, 3 stores, 6 product categories (Coffee, Tea, Bakery, Drinking Chocolate, Coffee Beans, Branded), and dozens of individual products, averaging about 24.9K customers served per month. The raw file had no revenue column and no time-intelligence columns, so those all had to be created during data preparation.

5 How It Was Built (Pipeline)



Every layer feeds the next: the raw CSV is cleaned once in Power Query, loaded into Power Pivot's in-memory model, measured with DAX, summarized in back-end pivot tables, and finally presented on a one-page executive dashboard connected by slicers and VBA reset buttons.

Although this project analyzes six months of sales data, the Power Query pipeline and reusable DAX measures allow it to scale to additional reporting periods with a simple refresh.

6 Data Preparation & Cleaning (Power Query)

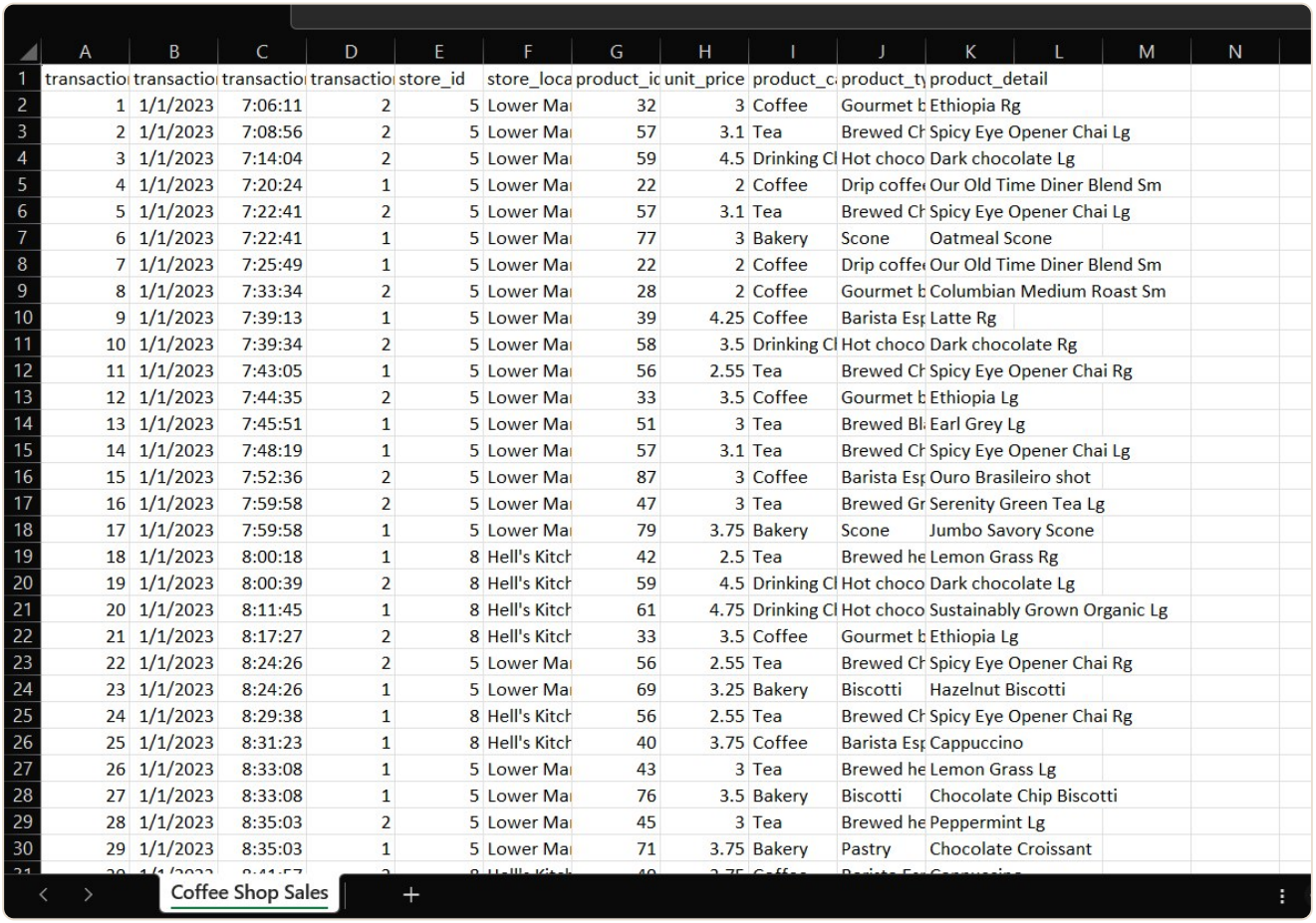
All cleaning and shaping were done in Power Query, so the whole process is repeatable. When new data arrives, one refresh reruns every step. The actual applied steps in my query:

1. **Source:** imported the CSV (11 columns, comma-delimited).
2. **Promoted Headers:** used the first row as column headers.
3. **Changed Type:** set correct data types for all 11 columns (whole numbers, date, time, decimal, text).
4. **Revenue Column:** added a calculated column: `transaction_qty × unit_price`, typed as decimal.

5. **Hour Column:** extracted the hour from `transaction_time` using `Time.Hour`, for hourly traffic analysis.
6. **DayName Column:** added the weekday name from `transaction_date` using `Date.DayOfWeekName`.
7. **MonthYear, MonthNumber, Month Name, Month:** added month columns in several formats ("Jan 2023" for chart labels, month numbers for correct sorting and time-based DAX).
8. **DayPart Column:** a conditional column grouping hours into Morning (6–11 AM), Afternoon (12–5 PM), and Evening (6–8 PM).
9. **Hour Label Column:** converted the numeric hour into readable labels like "9 AM" and "1 PM" for the charts.

SAMPLE M CODE: DAYPART LOGIC

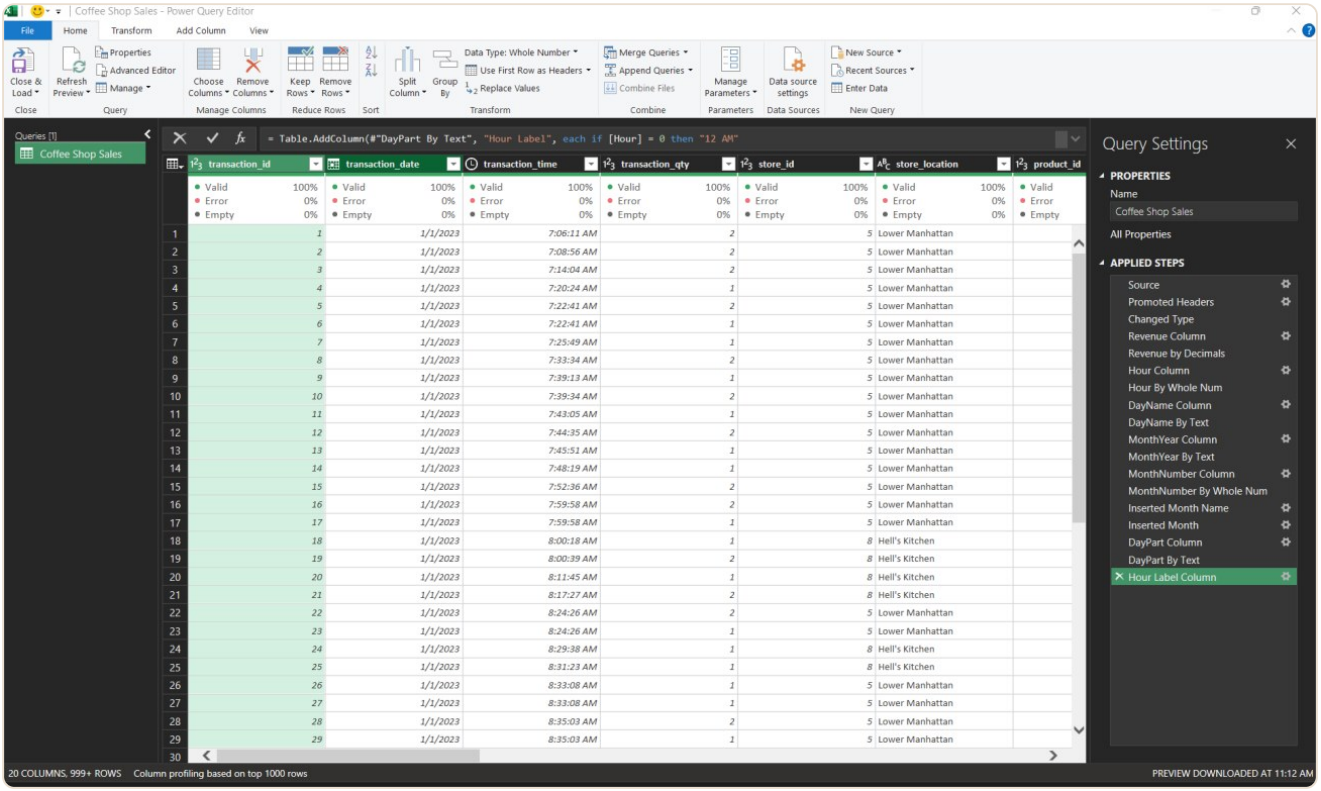
```
each if [Hour] >= 6  and [Hour] <= 11 then "Morning"
else if [Hour] >= 12 and [Hour] <= 17 then "Afternoon"
else if [Hour] >= 18 and [Hour] <= 20 then "Evening"
else "Other"
```



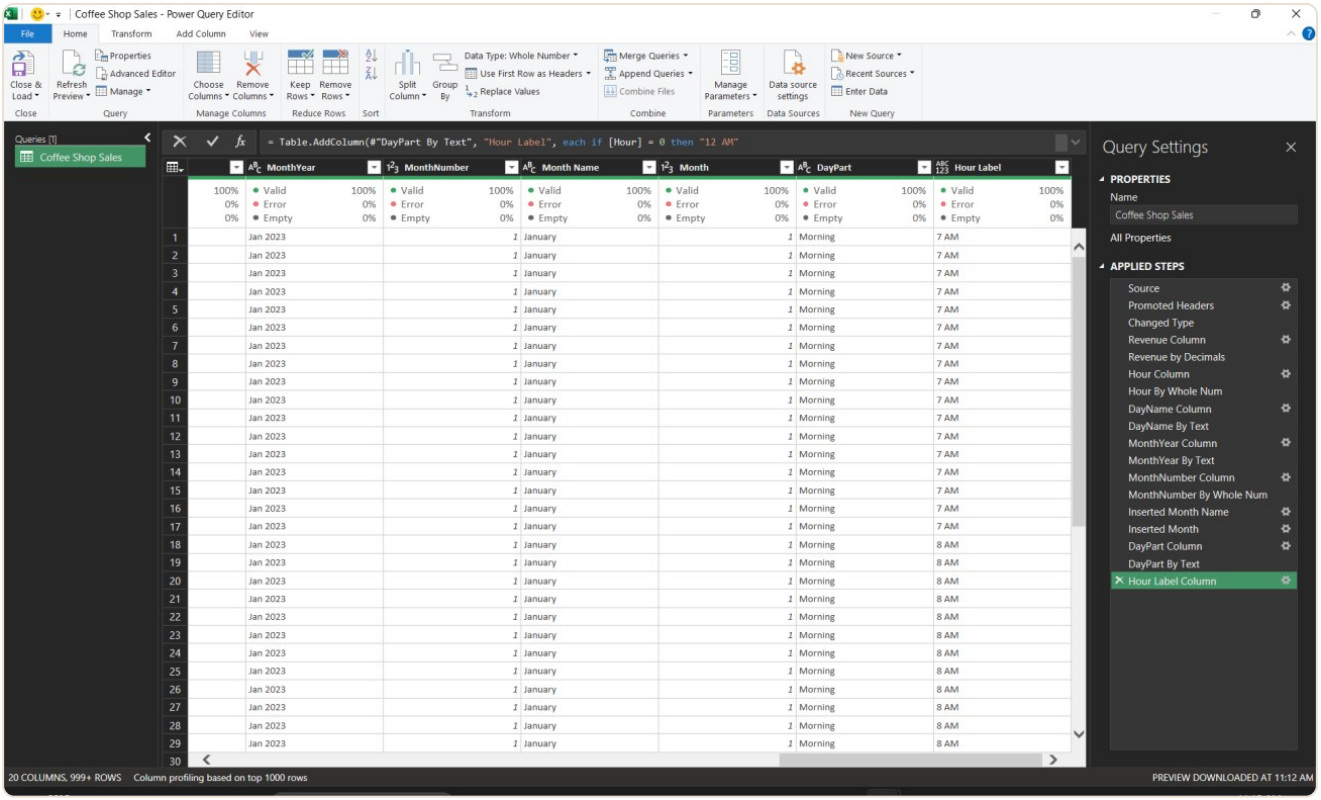
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	transactio	transactio	transactio	transactio	store_id	store_loca	product_ic	unit_price	product_c	product_t	product_detail			
2	1	1/1/2023	7:06:11	2	5	Lower Ma	32	3	Coffee	Gourmet b	Ethiopia Rg			
3	2	1/1/2023	7:08:56	2	5	Lower Ma	57	3.1	Tea	Brewed Ch	Spicy Eye Opener Chai Lg			
4	3	1/1/2023	7:14:04	2	5	Lower Ma	59	4.5	Drinking Ch	Hot choco	Dark chocolate Lg			
5	4	1/1/2023	7:20:24	1	5	Lower Ma	22	2	Coffee	Drip coffe	Our Old Time Diner Blend Sm			
6	5	1/1/2023	7:22:41	2	5	Lower Ma	57	3.1	Tea	Brewed Ch	Spicy Eye Opener Chai Lg			
7	6	1/1/2023	7:22:41	1	5	Lower Ma	77	3	Bakery	Scone	Oatmeal Scone			
8	7	1/1/2023	7:25:49	1	5	Lower Ma	22	2	Coffee	Drip coffe	Our Old Time Diner Blend Sm			
9	8	1/1/2023	7:33:34	2	5	Lower Ma	28	2	Coffee	Gourmet b	Columbian Medium Roast Sm			
10	9	1/1/2023	7:39:13	1	5	Lower Ma	39	4.25	Coffee	Barista Esp	Latte Rg			
11	10	1/1/2023	7:39:34	2	5	Lower Ma	58	3.5	Drinking Ch	Hot choco	Dark chocolate Rg			
12	11	1/1/2023	7:43:05	1	5	Lower Ma	56	2.55	Tea	Brewed Ch	Spicy Eye Opener Chai Rg			
13	12	1/1/2023	7:44:35	2	5	Lower Ma	33	3.5	Coffee	Gourmet b	Ethiopia Lg			
14	13	1/1/2023	7:45:51	1	5	Lower Ma	51	3	Tea	Brewed Bl	Earl Grey Lg			
15	14	1/1/2023	7:48:19	1	5	Lower Ma	57	3.1	Tea	Brewed Ch	Spicy Eye Opener Chai Lg			
16	15	1/1/2023	7:52:36	2	5	Lower Ma	87	3	Coffee	Barista Esp	Ouro Brasileiro shot			
17	16	1/1/2023	7:59:58	2	5	Lower Ma	47	3	Tea	Brewed Gr	Serenity Green Tea Lg			
18	17	1/1/2023	7:59:58	1	5	Lower Ma	79	3.75	Bakery	Scone	Jumbo Savory Scone			
19	18	1/1/2023	8:00:18	1	8	Hell's Kitch	42	2.5	Tea	Brewed he	Lemon Grass Rg			
20	19	1/1/2023	8:00:39	2	8	Hell's Kitch	59	4.5	Drinking Ch	Hot choco	Dark chocolate Lg			
21	20	1/1/2023	8:11:45	1	8	Hell's Kitch	61	4.75	Drinking Ch	Hot choco	Sustainably Grown Organic Lg			
22	21	1/1/2023	8:17:27	2	8	Hell's Kitch	33	3.5	Coffee	Gourmet b	Ethiopia Lg			
23	22	1/1/2023	8:24:26	2	5	Lower Ma	56	2.55	Tea	Brewed Ch	Spicy Eye Opener Chai Rg			
24	23	1/1/2023	8:24:26	1	5	Lower Ma	69	3.25	Bakery	Biscotti	Hazelnut Biscotti			
25	24	1/1/2023	8:29:38	1	8	Hell's Kitch	56	2.55	Tea	Brewed Ch	Spicy Eye Opener Chai Rg			
26	25	1/1/2023	8:31:23	1	8	Hell's Kitch	40	3.75	Coffee	Barista Esp	Cappuccino			
27	26	1/1/2023	8:33:08	1	5	Lower Ma	43	3	Tea	Brewed he	Lemon Grass Lg			
28	27	1/1/2023	8:33:08	1	5	Lower Ma	76	3.5	Bakery	Biscotti	Chocolate Chip Biscotti			
29	28	1/1/2023	8:35:03	2	5	Lower Ma	45	3	Tea	Brewed he	Peppermint Lg			
30	29	1/1/2023	8:35:03	1	5	Lower Ma	71	3.75	Bakery	Pastry	Chocolate Croissant			

Coffee Shop Sales

The raw `Coffee Shop Sales` dataset with 149,116 transaction-level rows before shaping.



Power Query Editor showing the applied steps that build the cleaned, reusable transformation.



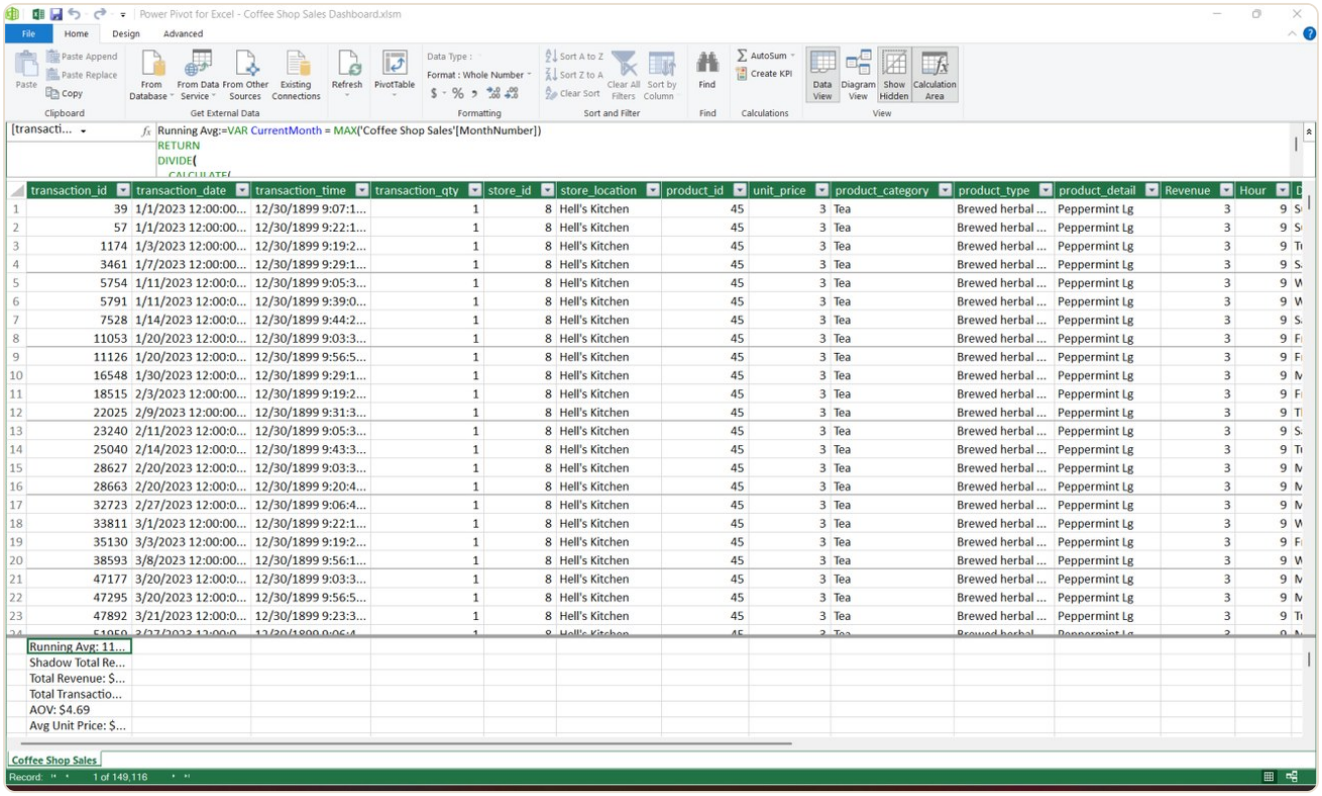
Calculated columns (DayPart, Month Name, Hour Label) generated once and refreshed on demand.

7 Data Model (Power Pivot)

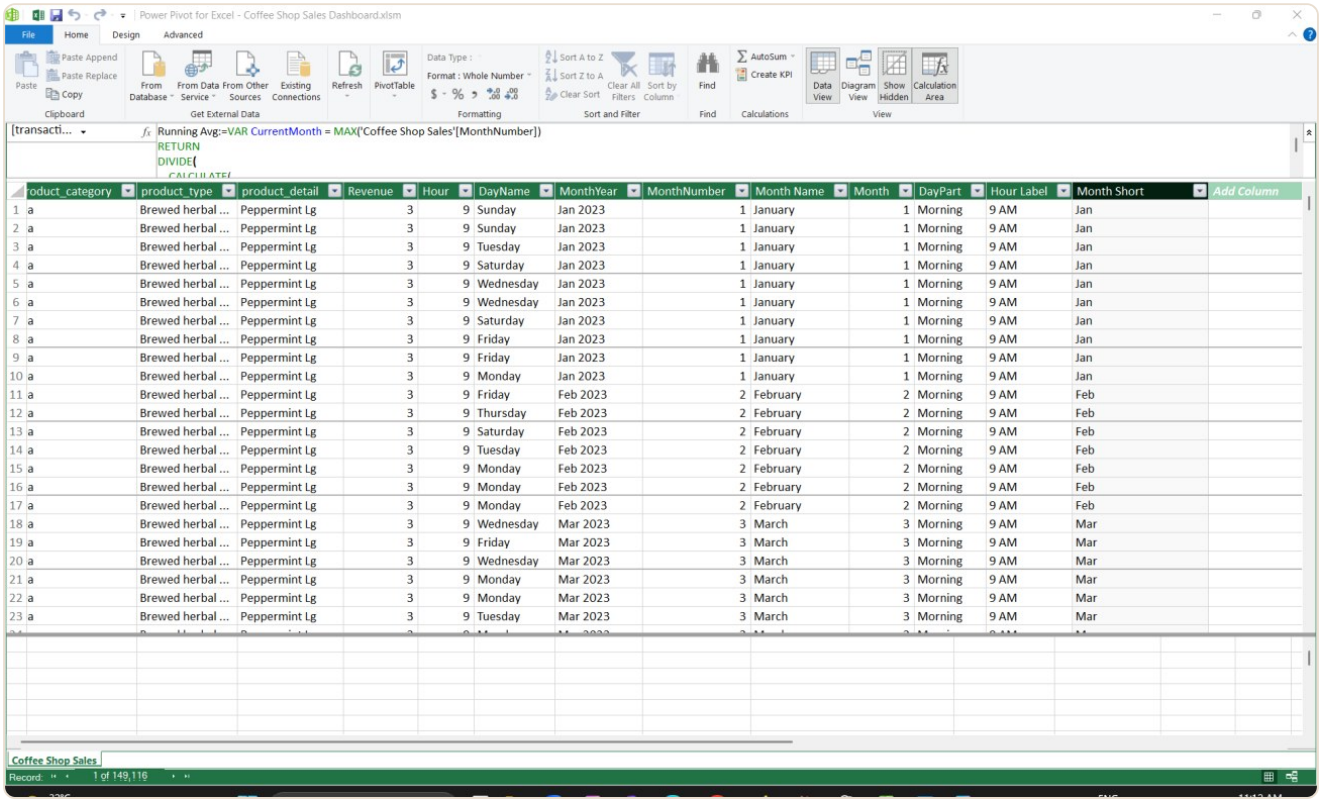
The cleaned table was loaded into Power Pivot's in-memory data model as a single table called `Coffee Shop Sales`, holding all 149K rows with the original columns plus the calculated columns from Power Query.

Why not a star schema?

A star schema separates a fact table (transactions) from dimension tables (products, stores, dates) connected by relationships. I chose not to build one here because the dataset is a single flat fact table where every attribute (store, product, category, date, and time) already lives on each transaction row. With only one table, splitting it into dimensions would have added relationships and lookup maintenance without changing any result. For a dataset of this size, a single denormalized table keeps the model simple, the DAX readable, and refreshes fast. In a larger, multi-source project (for example, adding cost data, targets per store, or a product master list), I would restructure this into a proper star schema.



Power Pivot's in-memory model: a single `Coffee Shop Sales` table with all measures attached.



The denormalized fact table with calculated columns feeding the DAX layer.

8 KPIs

The main KPI cards on the dashboard are:

KPI	Value	Why it matters
Total Revenue	**$698.81K**	The headline number; tracked against the $700K target (99.83% achieved).
MoM Growth	**+6.23%**	Shows momentum, not just totals.
Total Transactions	**149.1K**	Measures demand and foot traffic.
Average Order Value	**$4.69**	Tracked against a $5 target; shows spend per customer.
Peak Hour	**10 AM**	Directly drives staffing decisions.
Best-Selling Product	**Sustainably Grown Organic Lg**	Tells marketing what to promote.

Why these KPIs?

I picked KPIs that each lead to a decision, not just a number. Revenue and growth answer "how are we doing?"; transactions and AOV answer "is the problem traffic or spend per customer?"; and peak hour and best seller answer "what should we do about it?" Together they cover money, momentum, customers, and operations on one screen.

9 Calculated Measures (DAX)

I wrote 18 DAX measures in Power Pivot. The core ones:

```
Total Revenue      = SUM('Coffee Shop Sales'[Revenue])
Total Transactions = COUNTROWS('Coffee Shop Sales')
AOV                = DIVIDE([Total Revenue], [Total Transactions])
Avg Unit Price     = AVERAGE('Coffee Shop Sales'[unit_price])

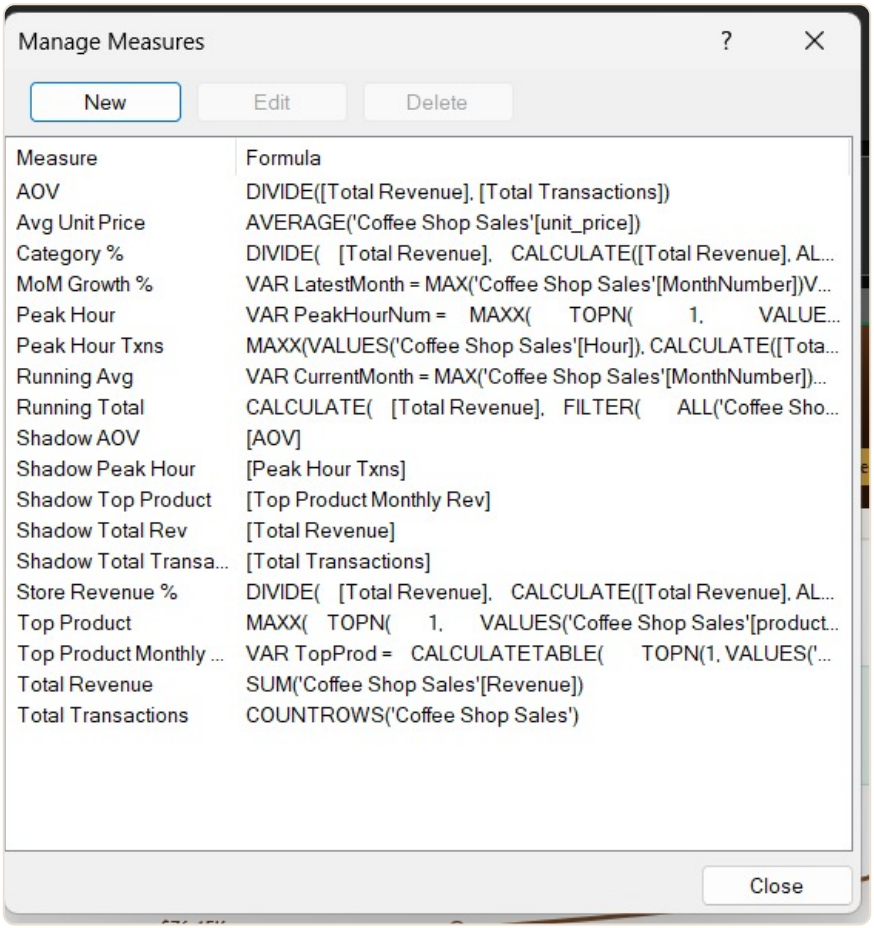
Category %      = DIVIDE([Total Revenue],
                    CALCULATE([Total Revenue],
                      ALL('Coffee Shop Sales'[product_category])))

Store Revenue % = DIVIDE([Total Revenue],
                    CALCULATE([Total Revenue],
                      ALL('Coffee Shop Sales'[store_location])))
```

The more advanced measures use variables (`VAR` / `RETURN`):

- **MoM Growth %:** captures the latest month with `MAX([MonthNumber])`, calculates current and previous month revenue with `CALCULATE`, then returns the percentage change with `DIVIDE`.
- **Running Avg:** `VAR CurrentMonth = MAX([MonthNumber])`, then divides the cumulative revenue (a `FILTER` over `ALL` month numbers up to the current one) by the number of months elapsed. This powers the dashed running-average line on the trend chart.
- **Running Total:** the cumulative revenue pattern on its own: `CALCULATE([Total Revenue], FILTER(ALL([MonthNumber]), [MonthNumber] <= MAX([MonthNumber])))`.
- **Peak Hour:** uses `TOPN(1, VALUES([Hour]), …)` to find the busiest hour by transaction count, then formats it into a readable label like "10 AM" with `IF` logic.
- **Peak Hour Txns:** `MAXX(VALUES([Hour]), CALCULATE([Total Transactions]))` returns the transaction count of that busiest hour.
- **Top Product:** `MAXX(TOPN(1, VALUES([product_detail]), CALCULATE([Total Revenue])), [product_detail])` returns the name of the highest-revenue product, and it re-evaluates under any slicer selection.
- **Top Product Monthly Rev:** a `CALCULATETABLE` + `TOPN` variable pattern that returns the top product's revenue for the selected period.
- **Shadow measures** (Shadow Total Rev, Shadow AOV, Shadow Peak Hour, Shadow Top Product, Shadow Total Transactions): lightweight aliases of the core measures, used by the back-end analysis pivots so the dashboard cards and the analysis layer always stay in sync.

Because these are reusable measures instead of hard-coded formulas, every pivot table and card recalculates consistently under any slicer combination.

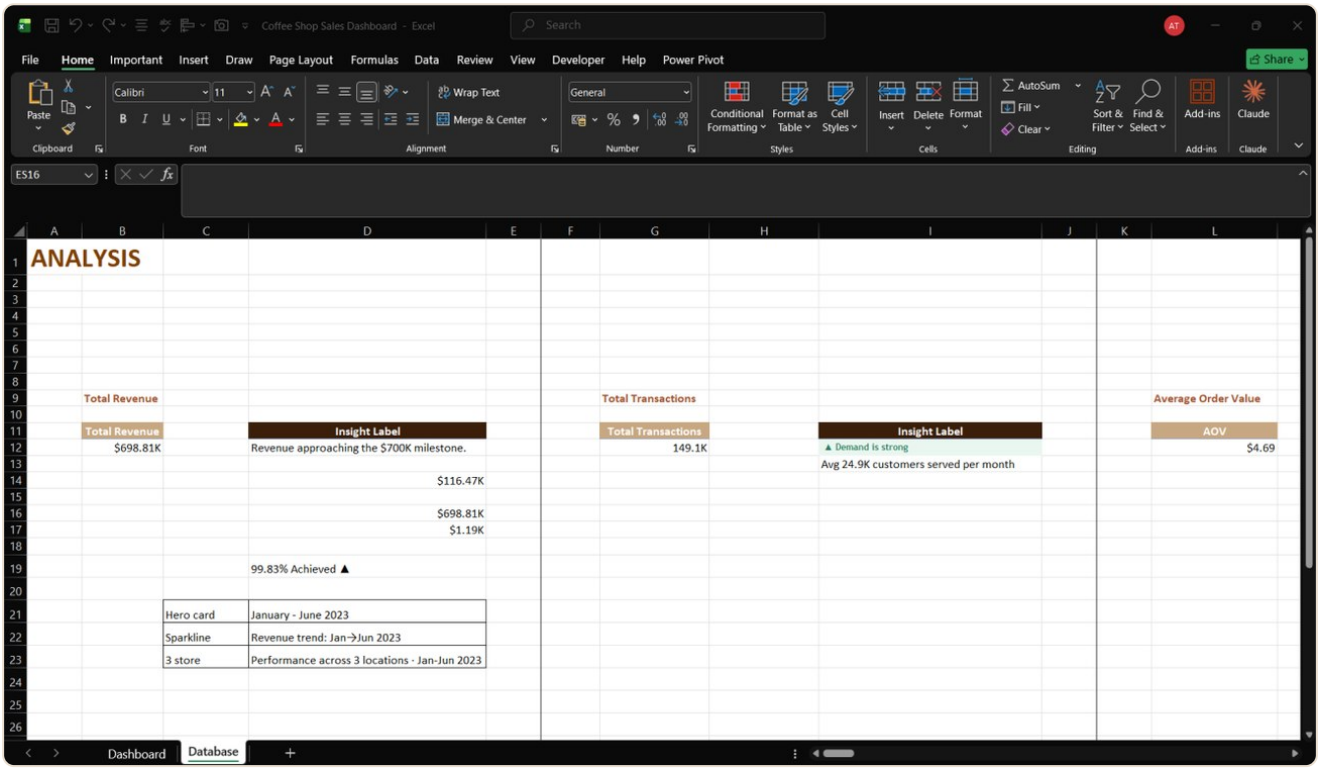


The Manage Measures dialog with all 18 reusable DAX measures driving every card and pivot.

10 Pivot Tables & Dashboard Connection

The dashboard does not sit directly on the data. Instead, there is a back-end Database sheet holding 20 pivot tables, each answering one question: monthly revenue, store revenue, hourly traffic, day-of-week revenue, category mix, top products, and the KPI values. The dashboard sheet then pulls from these pivots through cell links and formulas into the styled cards, charts, and insight labels.

Two slicers (Month Name and Store Location) are connected across the pivot tables through their slicer caches, so one click filters the entire page at once.



ANALYSIS

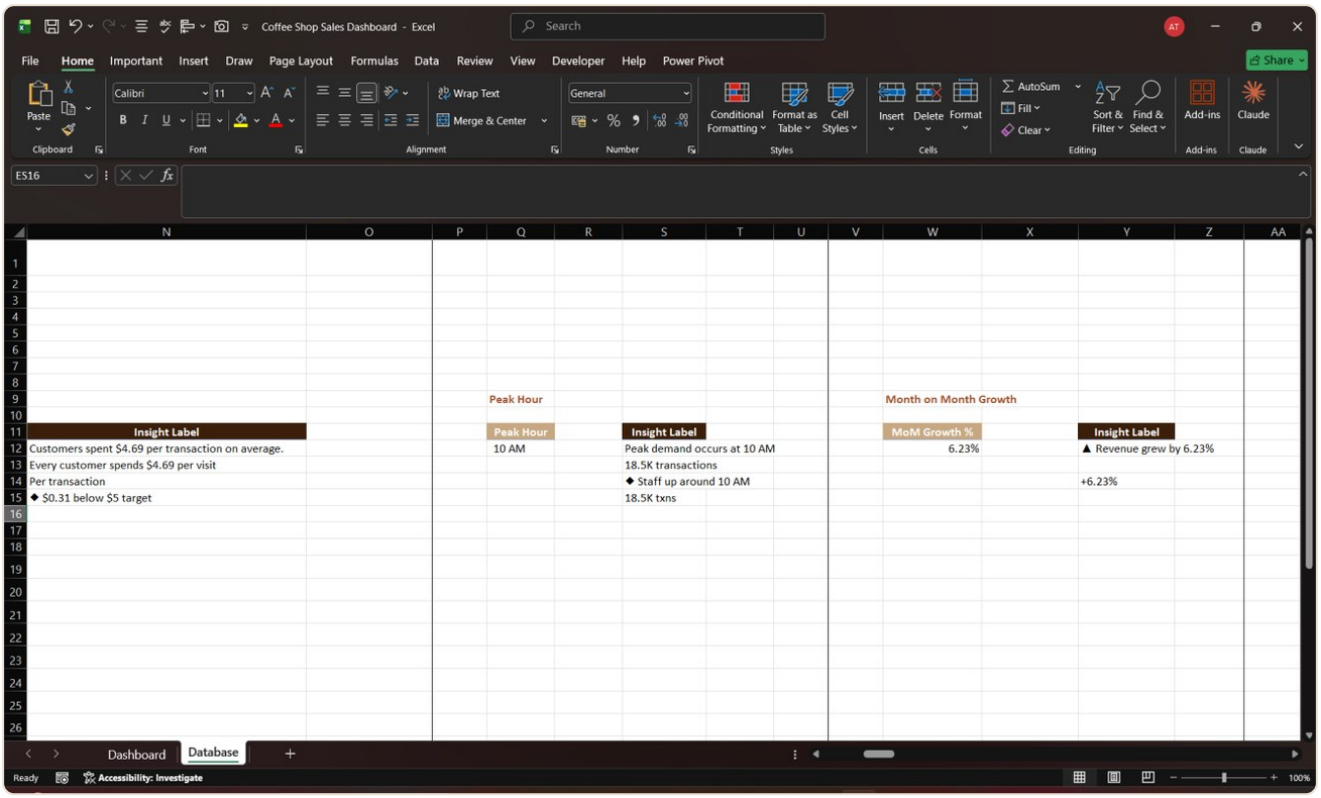
Total Revenue		
Total Revenue	Insight Label	
$698.81K	Revenue approaching the $700K milestone.	
		$116.47K
		$698.81K
		$1.19K
	99.83% Achieved ▲	

Hero card	January - June 2023
Sparkline	Revenue trend: Jan→Jun 2023
3 store	Performance across 3 locations · Jan-Jun 2023

Total Transactions		
Total Transactions	Insight Label	
149.1K	▲ Demand is strong	
	Avg 24.9K customers served per month	

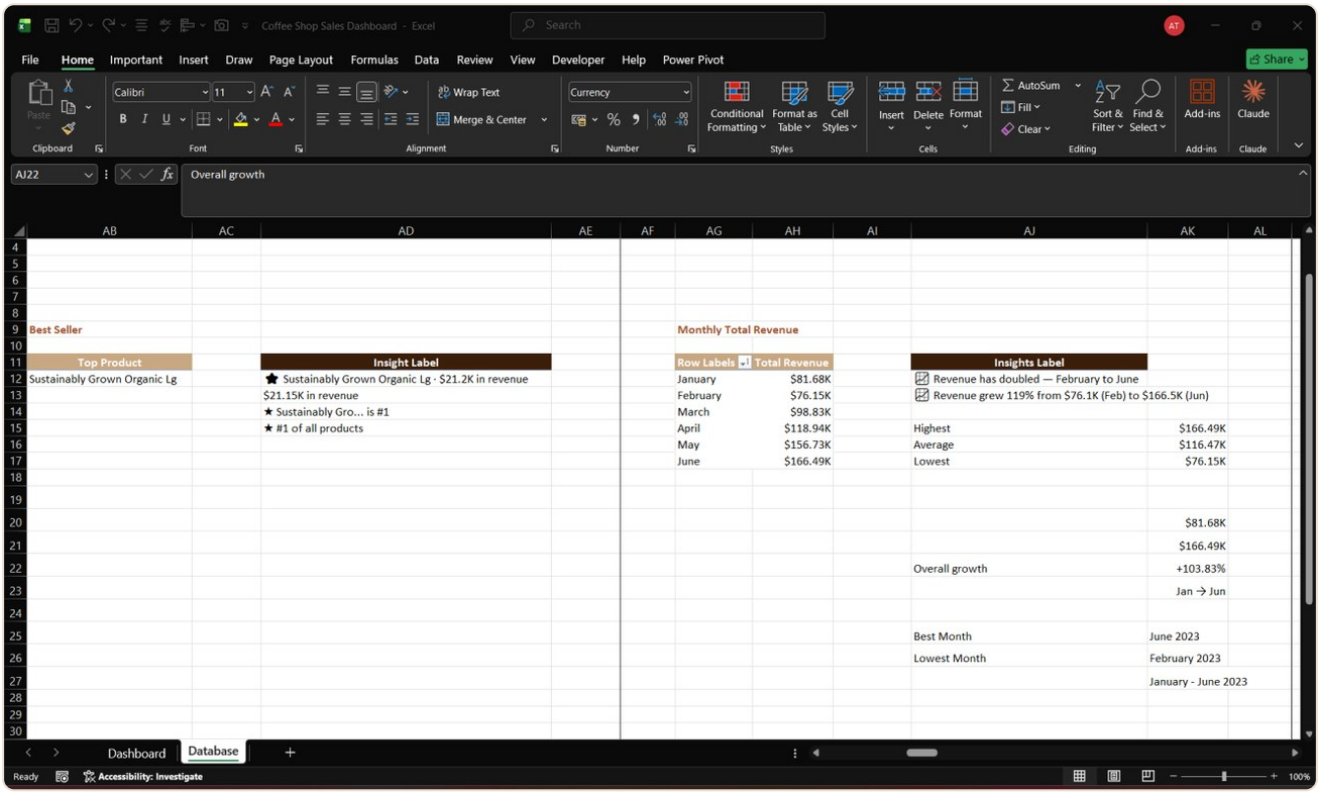
Average Order Value	
AOV	
$4.69	

Back-end Analysis layer: insight labels generated directly from the KPI calculations.

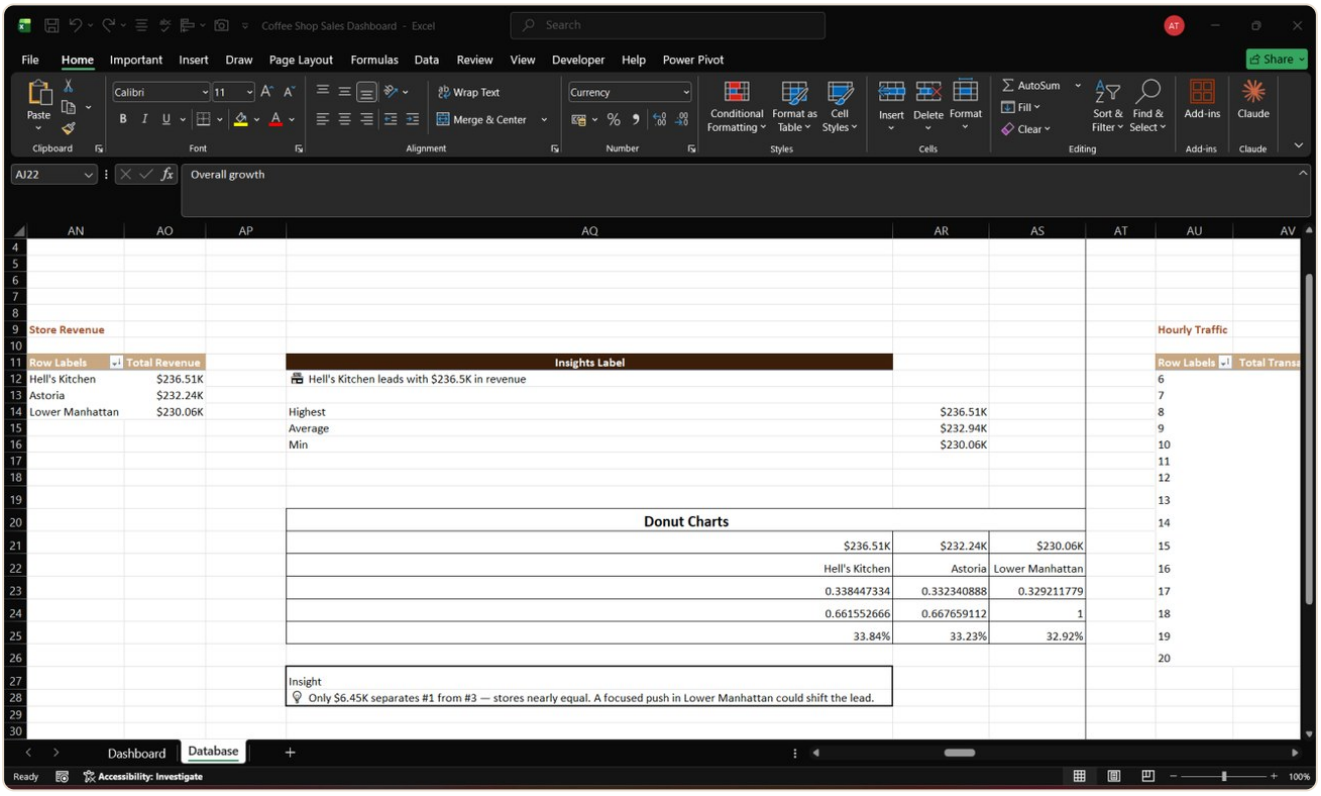


		Peak Hour		Month on Month Growth	
Insight Label		Peak Hour	Insight Label	MoM Growth %	Insight Label
Customers spent $4.69 per transaction on average.		10 AM	Peak demand occurs at 10 AM	6.23%	▲ Revenue grew by 6.23%
Every customer spends $4.69 per visit			18.5K transactions		
Per transaction			◆ Staff up around 10 AM		+6.23%
◆ $0.31 below $5 target			18.5K txns		

Peak-hour and month-on-month analysis feeding the dashboard's insight cards.



Monthly revenue and best-seller pivots with auto-generated narrative insights.



Store revenue and hourly-traffic pivots powering the donut charts and rankings.

11 VBA (Slicer Reset Buttons)

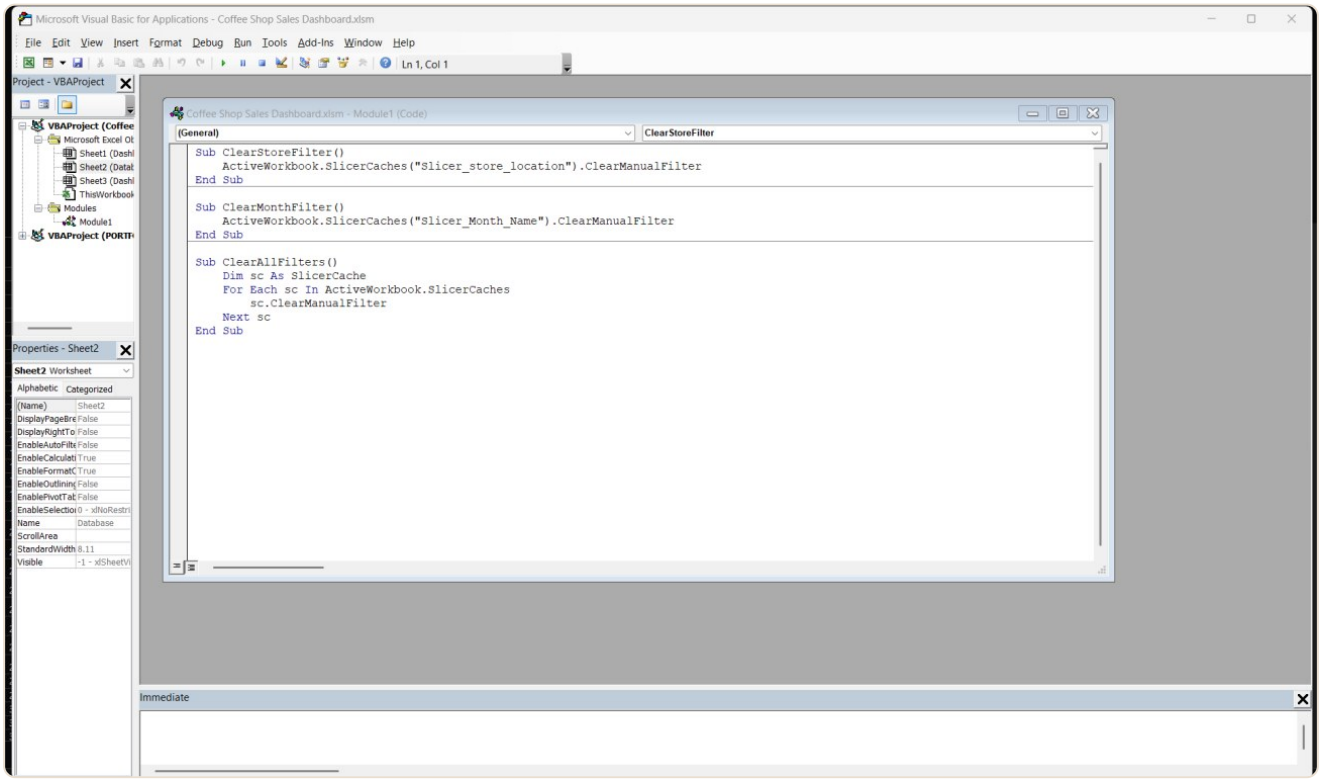
I wrote three macros so users can reset filters with one click instead of clearing each slicer manually:

```
Sub ClearStoreFilter()
    ActiveWorkbook.SlicerCaches("Slicer_store_location").ClearManualFilter
End Sub

Sub ClearMonthFilter()
    ActiveWorkbook.SlicerCaches("Slicer_Month_Name").ClearManualFilter
End Sub

Sub ClearAllFilters()
    Dim sc As SlicerCache
    For Each sc In ActiveWorkbook.SlicerCaches
        sc.ClearManualFilter
    Next sc
End Sub
```

These are assigned to the reset buttons on the dashboard, which makes the file feel more like an app than a spreadsheet.



The three reset macros in the VBA editor, wired to on-sheet buttons.

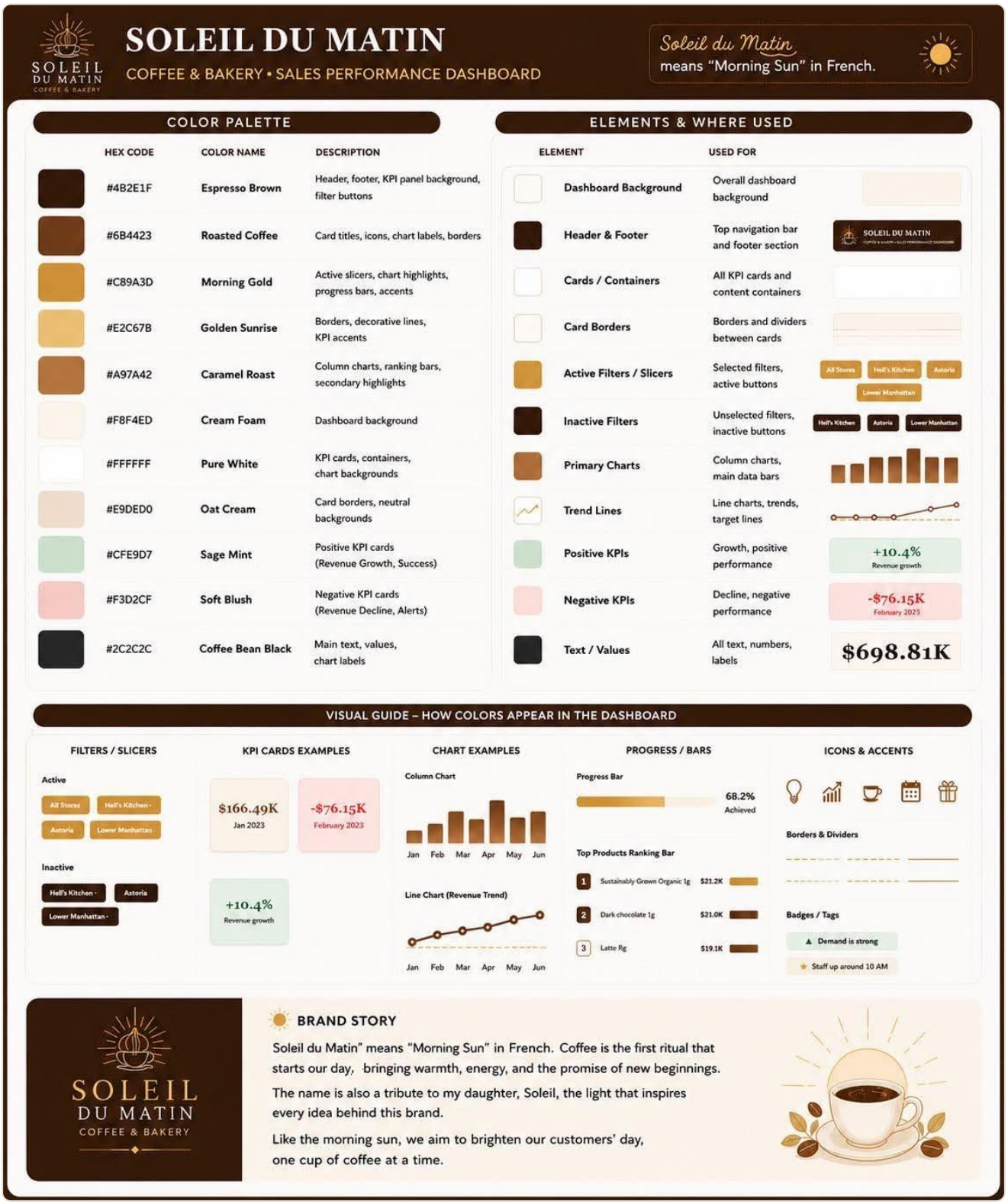
12 Dashboard Design Decisions

Why these colors? The warm browns, creams, and gold come straight from the Soleil du Matin logo, so the dashboard looks like it belongs to the brand. The palette also fits a coffee business naturally, and I reserved gold highlights for the most important values (like the peak hour bar) so the eye goes there first. Green, red, and neutral tags mark good, weak, and neutral results.

Why a line chart instead of an area chart? The monthly trend needs to show two series at once: actual revenue and the running average. Two overlapping areas would cover each other and make the chart harder to read. A solid line plus a dashed line keeps both visible, and the exact data labels stay clear.

Why these categories? The six product categories (Coffee, Tea, Bakery, Drinking Chocolate, Coffee Beans, Branded) match how the business organizes its menu and inventory, so the revenue mix maps directly to decisions the owner can act on, like which section of the menu to grow.

What business decisions could be made from this dashboard? Staffing levels by hour, whether to run a weekend promotion, which products to market as signature items, whether to shorten evening hours, and where pricing or quality changes carry the most risk. Each section of the dashboard connects to at least one of these calls.



The brand system: color palette, usage guide, and story, carried from logo to analytics.

13 Visualization Choices

Every visual on the dashboard was chosen to answer a specific business question, not for decoration. Here is the reasoning behind each one:

Element	Visual used	Why this visual	Business question
Total Revenue (Hero)	KPI card + progress bar + sparkline	Fastest way to read performance at a glance; the progress bar shows distance to target, the sparkline shows the trend without taking space.	Are we making enough, and are we close to target?
Monthly Revenue Trend	Line chart + running average	A line shows change over time clearly; the dashed running average smooths short-term swings so above/below-trend months stand out.	Is revenue increasing or decreasing?
Strategic Insights	Insight cards from KPI calculations	Executives don't always have time to read charts, so findings are turned into actions ("Revenue grew 104%, continue strategy").	What action should management take?
Total Transactions	KPI card + sparkline	Revenue alone doesn't explain performance; transactions show demand and traffic, and the sparkline shows the direction.	Are more customers visiting?
Average Order Value	KPI card + sparkline	Shows whether growth comes from higher spend per customer or just more customers.	Are customers spending enough?
Peak Hour	KPI card + sparkline	The sparkline adds context on transaction volume across the day, not just the single busiest hour.	When should we schedule more staff?
Best-Selling Product	KPI card + sparkline	Identifies the top product and shows whether its sales stay consistent over time.	Which product deserves more marketing?
Revenue by Store	3 KPI cards + radial charts	With only three stores, cards make values instantly readable; the radials emphasize each store's share while staying compact.	Which store performs best?
Transactions by Hour	Column chart (10 AM highlighted)	Columns make hour-to-hour comparison easy; the gold bar draws the eye straight to the busiest hour.	What time should we add staff?
Revenue by Category	Donut chart	The goal is composition, not trend, so a donut shows each category's share of total revenue.	Which categories earn the most?
Revenue by Day of Week	Column chart	Comparing and ranking days is easier with column heights than with a line.	Which day performs best?
When Do Customers Visit	Horizontal progress bars	Shows proportions (morning / afternoon / evening), not trends.	Which time of day draws the most customers?

Element	Visual used	Why this visual	Business question
Top 7 Products	Ranked table + data bars	Management needs exact values, not percentages; data bars make the gaps visible at a glance.	Which products generate the most revenue?
Key Takeaways	Insight cards	Summarize the whole dashboard into statements an executive can read in under a minute.	What should management remember?

Why sparklines on the KPI cards?

A KPI card shows the current value but not whether it's improving or declining. The sparkline adds historical direction without taking up space, so a reader sees both the number and where it's heading.

Design philosophy

Every visualization was chosen to answer a specific business question. Line charts are for trends over time, column charts compare values across categories or hours, KPI cards highlight single important metrics, donut charts show composition, progress bars show proportions or progress toward a goal, and ranked tables give detailed comparisons. The visual follows the question, not personal preference. The goal wasn't the most colorful dashboard, but one that supports faster, data-driven decisions.

14 Key Insights

- **Revenue is growing fast.** Sales went from $76.15K in February to $166.49K in June, a 103.83% increase, with 6.23% growth in the latest month. Total revenue reached $698.81K, which is 99.83% of the $700K target.
- **Mornings carry the business.** 54.82% of all transactions happen before noon, and 10 AM is the single busiest hour with 18.5K transactions. Evenings bring in only 9.52% of visits.
- **The three stores are almost tied.** Hell's Kitchen leads with $236.51K, but only about $6.5K separates first place from third. A small push at any store could change the ranking.
- **Coffee drives most of the revenue.** Coffee alone makes up 38.63% of sales, and together with tea (28.11%), drinks account for about two-thirds of revenue. Bakery items only contribute 11.78%.
- **Weekdays beat weekends.** Monday is the busiest day at $101.68K while Saturday is the weakest at $96.89K, a gap of $4.78K.
- **Two products lead the menu.** Sustainably Grown Organic Lg ($21.2K) and Dark Chocolate Lg ($21.0K) are only $146 apart at the top of the product ranking.
- **Average order value is slightly below target.** At $4.69 per transaction, it falls $0.31 short of the $5 goal.

15 Business Recommendations

1. **Add staff before the 10 AM rush.** Two extra staff starting at 9 AM would cover the morning window that brings in 55% of all traffic.
2. **Run a Saturday promotion or weekend loyalty reward.** Closing the $4.78K weekend gap could add around $10–15K in monthly revenue.
3. **Protect coffee quality and pricing.** Since coffee is 38.63% of sales, any drop in quality would hit revenue directly.

4. **Market the top two products as signature items.** Sustainably Grown Organic Lg and Dark Chocolate Lg are natural picks for a "Soleil Signature" lineup.
5. **Use bundles to raise the average order value past $5.** A morning coffee-and-pastry combo would also help the bakery category, which is currently underperforming.
6. **Consider shorter evening hours.** With less than 10% of traffic after 6 PM, cutting back evening operations could save labor costs without losing much revenue.

16 Technical Challenges

- **Building a running average in DAX.** Getting the running average to respect month order and still respond correctly to slicers took several iterations of the `VAR` + `FILTER` + `ALL` pattern before the dashed trend line behaved properly.
- **Creating a dynamic Top Product with `TOPN` and `MAXX`.** The best-seller card had to re-evaluate itself for any month or store selection instead of showing a fixed name, which meant combining `TOPN` inside `MAXX` rather than a simple lookup.
- **Designing slicers that update every KPI.** Twenty pivot tables, all the charts, and the insight labels are connected to the same two slicer caches, so one click filters the entire dashboard consistently.
- **Balancing executive readability with detailed analytics.** The hardest part was fitting deep analysis (hourly traffic, rankings, category share) on one page without it feeling crowded, which meant cutting anything that did not lead to a decision.
- **Performance optimization.** I used Power Pivot's in-memory data model and reusable DAX measures instead of thousands of worksheet formulas, which keeps slicer interactions responsive even with 149K rows.

17 Impact / Value

- Turned 149,116 raw transactions into a single page that management can read in seconds, instead of digging through spreadsheets.
- Found a concrete revenue opportunity (the Saturday promo, worth an estimated $10–15K per month) and a staffing plan backed by real traffic data.
- Built the file to be reusable: the Power Query pipeline, DAX measures, and slicer-connected pivots mean it can refresh for future periods without manual rework.
- Covers the full Excel BI process from start to finish: cleaning data in Power Query, modeling in Power Pivot, writing 18 DAX measures, building 20 back-end pivot tables, adding VBA automation, and presenting everything in a clear, executive-level design.
- Includes original brand work as well: the concept, name, and logo were all created from scratch and carried through the whole dashboard.

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